ENAE 694 ● FALL 2026

Homework 3

Antenna Gain and Mission Trades

Assigned: September 28, 2026 Due: October 12, 2026 Value: 100 points

Show your calculations and units. Calculate gain directly in dB. Use D in meters and f in GHz. Use the supplied constants, retain intermediate precision, and round final answers to two decimal places. Set your calculator to degrees. You may revise and resubmit until correct; there are no late penalties.

1 Antenna Gain and Beamwidth — 30 Points

A circular spacecraft reflector antenna operates at f = 7.50 GHz, with diameter D = 0.80 m. Calculate peak gain directly from diameter and frequency using the formula below, which assumes aperture efficiency 0.55.

a. Calculate peak antenna gain [G] in dBi and full half-power beamwidth (HPBW) in degrees. [10 points]

b. Double the diameter while keeping frequency and efficiency fixed. Find the change in gain, the new gain, and the new HPBW. [10 points]

c. Return to the original antenna. Double the frequency from 7.50 to 15.00 GHz while keeping diameter and efficiency fixed. Find the change in gain and the new HPBW. Explain why this gain increase alone does not establish the change in received power for the complete link. [10 points]

Use D in meters and f in GHz. The gain constant includes efficiency 0.55 and all unit conversions; do not apply efficiency again. HPBW is the full width in degrees. Its reflector rule is approximate and depends on illumination, not efficiency alone.

$$[G] = 17.81 + 20 \log_{10}(D) + 20 \log_{10}(f) \qquad HPBW \approx 21 / (f \ D) \ degrees$$

2 Pointing and Polarization Loss — 30 Points

A different link has aligned, polarization-matched received power [P_{r0}] = −118.00 dBW. The receiver requires at least −123.50 dBW. Transmit HPBW is 3.50° with a pointing error of 0.875°. Receive HPBW is 1.40° with a pointing error of 0.350°. Both antennas use linear polarization; their polarization axes differ by 45° about the line of sight. All other losses are already included in [P_{r0}].

a. Calculate the transmit and receive pointing losses separately. [10 points]

b. Calculate polarization loss, actual received power, and margin above the receiver threshold. Does the link meet the threshold? [10 points]

c. Change only the polarization-axis difference to 60°. Recalculate received power and margin. Explain why correcting polarization alignment can restore this link without changing antenna gain. [10 points]

Use the following near-boresight approximation and ideal linear-polarization model. Enter θ and full HPBW in the same angular units. Record losses as positive dB values and subtract them once. The pointing approximation does not model sidelobes or nulls.

$$[L_{point}] = 12 \ (\theta \ / \ HPBW)^2 \qquad [L_{pol}] = -10 \log_{10}(\cos^2 \psi)$$

$$[P_r] = [P_{r0}] - [L_{pt}] - [L_{pr}] - [L_{pol}] \qquad Margin = [P_r] - Threshold$$

3 Antenna Pointing During a Pass — 20 Points

A spacecraft must keep its science instrument pointed at nadir while communicating continuously throughout a 12-minute overhead pass. Its fixed antenna also points at nadir. At the midpoint (6 minutes), the station lies on antenna boresight: gain is at its peak, range is shortest, and link margin is +5 dB. Toward either end, the station moves off boresight, gain decreases, and range increases. Use the interval approximations below; all other link-budget terms remain unchanged. The link is usable when margin is zero or greater.

Time After Start of Pass	Fixed Antenna Gain Relative to Peak	Additional Path Loss
0–2 min	−8 dB	4 dB
2–5 min	−2 dB	1 dB
5–7 min	0 dB	0 dB
7–10 min	−2 dB	1 dB
10–12 min	−8 dB	4 dB

a. For the fixed antenna, calculate margin in each interval. Identify when the link is usable. Does it meet the continuous-communication requirement? [8 points]

b. Put the same antenna on a gimbal that tracks the station throughout the pass while the spacecraft remains nadir-pointed. Assume ideal tracking, adequate gimbal travel, and no added RF loss: antenna gain stays at its peak, but the additional path loss remains as listed. Calculate margin in each interval. Does this configuration meet the requirement? [8 points]

c. Explain why a gimbal is needed for this antenna to meet both requirements. Why would rotating the whole spacecraft toward the station not be an acceptable solution? [4 points]

4 Antenna Mass and Pointing Trade — 20 Points

Continue the spacecraft scenario from Problem 3. The spacecraft must maintain nadir-pointed science observations and communicate throughout the full pass. Both antenna options below have the RF performance used in Problem 3. The maximum installed antenna-system mass is 5.0 kg.

Option	Complete Installed Mass	Pointing Capability
A — Fixed antenna	2.0 kg	Requires spacecraft rotation to point at the station.
B — Gimbaled antenna	6.0 kg	Tracks the station while the spacecraft maintains its science attitude.

The stated masses include the aperture, feed, supports, and any deployment or pointing mechanisms. These are supplied design assumptions, not specifications for a named mission.

a. Using your results from Problem 3, identify which option meets the communications and science requirements. Compare each option with the mass limit. Does either meet all requirements as proposed? [8 points]

b. Recommend one design or requirement change that makes the mission feasible. State its cost or consequence, including any mass reallocation, hardware redesign, or reduction in communications or science performance. [6 points]

c. Explain why antenna mass location matters to spacecraft slewing and pointing, even when two designs have the same total mass. [6 points]

For Problem 3: margin = 5 dB + gain relative to peak − additional path loss. Additional path loss is relative to the pass midpoint. A gimbal changes pointing, not range. No outside research is required.